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Ionic Equilibrium

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MCQ (Single Correct Answer)	86 questions
Answer key	at the end

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MCQ (Single Correct Answer) 86 questions

Q1. RE-NEET 2026

The correct order of solubility of the given salts in water at $298\ \mathrm{K}$ is

Salt	K_{sp} at 298 K
$AgBr$	5.0×10^{-13}
$Zn(OH)_2$	1.0×10^{-15}
Hg_2Cl_2	1.3×10^{-18}

(A) $Zn(OH)_2 > AgBr > Hg_2Cl_2$
(B) $Hg_2Cl_2 > Zn(OH)_2 > AgBr$
(C) $AgBr > Zn(OH)_2 > Hg_2Cl_2$
(D) $Hg_2Cl_2 > AgBr > Zn(OH)_2$

Open solution

Q2. NEET 2026

In a qualitative analysis, Bi^{3+} is detected by appearance of precipitate of $BiO(OH)(s)$. Calculate pH when the following equilibrium exists at 298 K .

$BiO(OH)(s) \rightleftharpoons BiO^{+}(aq) + OH^{-}(aq), K = 4 \times 10^{-10}$

(Given : $\log 2 = 0.3010$)

(A) 8.714 (B) 4.699 (C) 5.286 (D) 9.301

Open solution

Q3. NEET 2026

At 298 K , a certain buffer solution contains equal concentrations of $X^{-}$and HX, K_b for $X^{-}$is 10^{-10}. What is the pH of this buffer solution?

(A) 2 (B) 4 (C) 6 (D) 10

Open solution

Q4. NEET 2026

Phenolphthalein is used as an indicator for the titration of sodium hydroxide solution against a standard solution of oxalic acid. The colour change that is observed at an alkaline pH close to the equivalence point during this titration is:

(A) pinkish red to yellow
(B) yellow to pinkish red
(C) pink to colourless
(D) colourless to pink

Open solution

Q5. NEET 2025

If the molar conductivity (Λ_m) of a $0.050\ \mathrm{mol\ L^{-1}}$ solution of a monobasic weak acid is $90\ \mathrm{S\ cm^2\ mol^{-1}}$, its extent (degree) of dissociation will be

[Assume $\Lambda^{\circ}_{+} = 349.6\ \mathrm{S\ cm^2\ mol^{-1}}$ and $\Lambda^{\circ}_{-} = 50.4\ \mathrm{S\ cm^2\ mol^{-1}}$.]

(A) 0.225 (B) 0.215 (C) 0.115 (D) 0.125

Open solution

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Q6. NEET 2025

Phosphoric acid ionizes in three steps with their ionization constant values K_{a_1}, K_{a_2} and K_{a_3}, respectively, while K is the overall ionization constant. Which of the following statements are true?

A. $\log \mathrm{K} = \log \mathrm{K}_{a_1} + \log \mathrm{K}_{a_2} + \log \mathrm{K}_{a_3}$

B. $\mathrm{H_3PO_4}$ is a stronger acid than $\mathrm{H_2PO_4^-}$ and $\mathrm{HPO_4^{2-}}$

C. $K_{a_1} > K_{a_2} > K_{a_3}$

D. $K_{a_1} = \frac{K_{a_3} + K_{a_2}}{2}$

Choose the correct answer from the options given below :

(A) B, C and D only (B) A, B and C only (C) A and B only (D) A and C only

Open solution

Q7. NEET 2024 (Re-Examination)

Which indicator is used in the titration of sodium hydroxide against oxalic acid and what is the colour change at the end point?

(A) Phenolphthalein, pink to yellow

(B) Alkaline $KMnO_4$, colourless to pink

(C) Phenolphthalein, colourless to pink

(D) Methyl orange, yellow to pinkish red colour

Open solution

Q8. NEET 2024 (Re-Examination)

The ratio of solubility of AgCl in 0.1 M KCl solution to the solubility of AgCl in water is:

(Given : Solubility product of AgCl = 10^{-10})

(A) 10^{-4} (B) 10^{-6} (C) 10^{-9} (D) 10^{-5}

Open solution

Q9. NEET 2023 Manipur

An acidic buffer is prepared by mixing :

(A) weak acid and it's salt with strong base

(B) equal volumes of equimolar solutions of weak acid and weak base

(C) strong acid and it's salt with strong base

(D) strong acid and it's salt with weak base (The pK_a of acid = pK_b of the base)

Open solution

Q10. NEET 2022 Phase 2

0.01 M acetic acid solution is 1% ionised, then pH of this acetic acid solution is :

(A) 1 (B) 3 (C) 2 (D) 4

Open solution

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Q11. NEET 2022 Phase 1

The pH of the solution containing 50 mL each of 0.10 M sodium acetate and 0.01 M acetic acid is [Given pK_a of CH_3COOH = 4.57]

(A) 5.57 (B) 3.57 (C) 4.57 (D) 2.57

Open solution

Q12. NEET 2021

The pK_b of dimethyl amine and pK_a of acetic acid are 3.27 and 4.77 respectively at T (K). The correct option for the pH of dimethyl ammonium acetate solution is :

(A) 6.25 (B) 8.50 (C) 5.50 (D) 7.75

Open solution

Q13. NEET 2020 Phase 2

Which among the following salt solutions is basic in nature ?

(A) Sodium acetate (B) Ammonium chloride

(C) Ammonium sulphate (D) Ammonium nitrate

Open solution

Q14. NEET 2020 Phase 2

The solubility product for a salt of the type AB is 4×10^{-8}. What is the molarity of its standard solution ?

(A) 4×10^{-4} mol/L (B) 2×10^{-4} mol/L (C) 16×10^{-16} mol/L (D) 2×10^{-16} mol/L

Open solution

Q15. NEET 2020 Phase 1

Find out the solubility of $Ni(OH)_2$ in 0.1M NaOH. Given that the ionic product of $Ni(OH)_2$ is 2×10^{-15}.

(A) 2×10^{-8} M (B) 1×10^{-13} M (C) 1×10^{8} M (D) 2×10^{-13} M

Open solution

Q16. NEET 2019 (Odisha)

The pH of $0.01\ \mathrm{M}\ \mathrm{NaOH}\ (\mathrm{aq})$ solution will be

(A) 9 (B) 7.01 (C) 2 (D) 12

Open solution

Q17. NEET 2019 (Odisha)

Which of the following **cannot** act both as Bronsted acid and as Bronsted base?

(A) $\mathrm{HSO_4^-}$ (B) $\mathrm{HCO_3^-}$ (C) $\mathrm{NH_3}$ (D) HCl

Open solution

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Q18. NEET 2019 (Odisha)

The molar solubility of CaF_2 ($K_{sp} = 5.3 \times 10^{-11}$) in 0.1 M solution of NaF will be

(A) 5.3×10^{-10} mol L^{-1} (B) 5.3×10^{-11} mol L^{-1}

(C) 5.3×10^{-8} mol L^{-1} (D) 5.3×10^{-9} mol L^{-1}

Open solution

Q19. NEET 2019

Conjugate base for Bronsted acids H_2O and HF are :

(A) OH^- and F^-, respectively (B) H_3O^+ and H_2F^+, respectively

(C) OH^- and H_2F^+, respectively (D) H_3O^+ and F^-, respectively

Open solution

Q20. NEET 2019

pH of a saturated solution of $Ca(OH)_2$ is 9. The solubility product (K_{sp}) of $Ca(OH)_2$ is :

(A) 0.125×10^{-15} (B) 0.5×10^{-10} (C) 0.5×10^{-15} (D) 0.25×10^{-10}

Open solution

Q21. NEET 2019

Which will make basic buffer?

(A) 100 mL of 0.1 M HCl + 200 mL of 0.1 M NH_4OH

(B) 100 mL of 0.1 M HCl + 100 mL of 0.1 M NH_4OH

(C) 50 mL of 0.1 M NaOH + 25 mL of 0.1 M CH_3COOH

(D) 100 mL of 0.1 M CH_3COOH + 100 mL of 0.1 M NaOH

Open solution

Q22. NEET 2018

Following solutions were prepared by mixing different volumes of NaOH and HCl of different concentrations :

A. 60 mL $\frac{M}{10}$ HCl + 40 mL $\frac{M}{10}$ NaOH

B. 55 mL $\frac{M}{10}$ HCl + 45 mL $\frac{M}{10}$ NaOH

C. 75 mL $\frac{M}{5}$ HCl + 25 mL $\frac{M}{5}$ NaOH

D. 100 mL $\frac{M}{10}$ HCl + 100 mL $\frac{M}{10}$ NaOH

pH of which one of them will be equal to 1?

(A) B (B) A (C) D (D) C

Open solution

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Q23. NEET 2018

The solubility of $BaSO_4$ in water is 2.42×10^{-3} g L^{-1} at 298 K. The value of its solubility product (K_{sp}) will be (Given molar mass of $BaSO_4$ = 233 g mol^{-1})

(A) 1.08×10^{-10} mol^2 L^{-2}
(B) 1.08×10^{-12} mol^2 L^{-2}
(C) 1.08×10^{-14} mol^2 L^{-2}
(D) 1.08×10^{-8} mol^2 L^{-2}

Open solution

Q24. NEET 2017

Concentration of the Ag^+ ions in a saturated solution of $Ag_2C_2O_4$ is 2.2×10^{-4} mol L^{-1}. Solubility product of $Ag_2C_2O_4$ is

(A) 2.66×10^{-12} (B) 4.5×10^{-11} (C) 5.3×10^{-12} (D) 2.42×10^{-8}

Open solution

Q25. NEET 2016 Phase 2

The percentage of pyridine (C_5H_5N) that forms pyridinium ion ($C_5H_5N^+H$) in a 0.10 M aqueous pyridine solution (K_b for C_5H_5N = 1.7×10^{-9}) is

(A) 0.0060% (B) 0.013% (C) 0.77% (D) 1.6%

Open solution

Q26. NEET 2016 Phase 2

The solubility of $AgCl_{(s)}$ with solubility product 1.6×10^{-10} in 0.1 M NaCl solution would be

(A) 1.26×10^{-5} M (B) 1.6×10^{-9} M (C) 1.6×10^{-11} M (D) zero

Open solution

Q27. NEET 2016 Phase 2

Which of the following fluro-compounds is most likely to behave as a Lewis base ?

(A) BF_3 (B) PF_3 (C) CF_4 (D) SiF_4

Open solution

Q28. NEET 2016 Phase 1

MY and NY_3, two nearly insoluble salts, have the same K_{sp} values of 6.2×10^{-13} at room temperature. Which statement would be true in regard to MY and NY_3?

(A) The salts MY and NY_3 are more soluble in 0.5 M KY than in pure water.
(B) The addition of the salt of KY to solution of MY and NY_3 will have no effect on their solubilities.
(C) The molar solubilities of MY and NY_3 in water are identical.
(D) The molar solubility of MY in water is less than that of NY_3.

Open solution

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Q29. AIPMT 2015

What is the pH of the resulting solution when equal volumes of 0.1 M NaOH and 0.01 M HCl are mixed?

(A) 2.0 (B) 7.0 (C) 1.04 (D) 12.65

Open solution

Q30. AIPMT 2015

Which one of the following pairs of solution is not an acidic buffer ?

(A) CH_3COOH and CH_3COONa (B) H_2CO_3 and Na_2CO_3

(C) H_3PO_4 and Na_3PO_4 (D) $HClO_4$ and $NaClO_4$

Open solution

Q31. AIPMT 2015 Cancelled Paper

The K_{sp} of Ag_2CrO_4, AgCl, AgBr and AgI are respectively, 1.1×10^{-12}, 1.8×10^{-10}, 5.0×10^{-13}, 8.3×10^{-17}. Which one of the following salts will precipitate last if $AgNO_3$ solution is added to the solution containing equal moles of NaCl, NaBr, NaI and Na_2CrO_4?

(A) AgBr (B) Ag_2CrO_4 (C) AgI (D) AgCl

Open solution

Q32. AIPMT 2014

Which of the following salts will give highest pH in water?

(A) KCl (B) NaCl (C) Na_2CO_3 (D) $CuSO_4$

Open solution

Q33. NEET 2013 (Karnataka)

Accumulation of lactic acid ($HC_3H_5O_3$), a monobasic acid in tissues leads to pain and a feeling of fatigue. In a 0.10 M aqueous solution, lactic acid is 3.7% dissociated. The value of dissociation constant, K_a, for this acid will be

(A) 1.4×10^{-5} (B) 1.4×10^{-4} (C) 3.7×10^{-4} (D) 2.8×10^{-4}

Open solution

Q34. NEET 2013 (Karnataka)

At 100°C the K_w of water is 55 times its value at 25°C. What will be the pH of neutral solution? (log 55 = 1.74)

(A) 7.00 (B) 7.87 (C) 5.13 (D) 6.13

Open solution

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Q35. NEET 2013 (Karnataka)

The values of K_{sp} of $CaCO_3$ and CaC_2O_4 are 4.7×10^{-9} and 1.3×10^{-9} respectively at 25°C. If the mixture of these two is washed with water, what is the concentration of Ca^{2+} ions in water ?

(A) 5.831×10^{-5} M
(B) 6.856×10^{-5} M
(C) 3.606×10^{-5} M
(D) 7.746×10^{-5} M

Open solution

Q36. NEET 2013 (Karnataka)

The dissociation constant of weak acid is 1×10^{-4}. In order to prepare a buffer solution with a pH = 5, the [Salt]/[Acid] ratio should be

(A) 4 : 5
(B) 10 : 1
(C) 5 : 4
(D) 1 : 10

Open solution

Q37. NEET 2013

$KMnO_4$ can be prepared from K_2MnO_4 as per the reaction,

$$3MnO_4^{2-} + 2H_2O \rightleftharpoons 2MnO_4^- + MnO^2 + 4OH^-$$

The reaction can go to completion by removing OH^- ions by adding

(A) CO_2
(B) SO_2
(C) HCl
(D) KOH

Open solution

Q38. NEET 2013

Which of these is least likely to act as a Lewis base ?

(A) BF_3
(B) PF_3
(C) CO
(D) F^-

Open solution

Q39. AIPMT 2012 Prelims

pH of a saturated solution of $Ba(OH)_2$ is 12. The value of solubility product (K_{sp}) of $Ba(OH)_2$ is

(A) 3.3×10^{-7}
(B) 5.0×10^{-7}
(C) 4.0×10^{-6}
(D) 5.0×10^{-6}

Open solution

Q40. AIPMT 2012 Prelims

Equimolar solutions of the following substances were prepared separately. Which one of these will record the highest pH value ?

(A) $BaCl_2$
(B) $AlCl_3$
(C) LiCl
(D) $BeCl_2$

Open solution

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Q41. AIPMT 2012 Prelims

Buffer solutions have constant acidity and alkalinity because

(A) these give unionised acid or base on reaction with added acid or alkali

(B) acids and alkalies in these solutions are shielded from attack by other ions

(C) they have large excess of H^+ or OH^- ions

(D) they have fixed value of pH

Open solution

Q42. AIPMT 2011 Mains

In qualitative analysis, the metals of group I can be separated from other ions by precipitating them as chloride salts. A solution initially contains Ag^+ and pb^{2+} at a concentration of 0.10 M. Aqueous HCl is added to this solution until the Cl^- concentration is 0.10 M. What will the concentrations of Ag^+ and Pb^{2+} be at equilibrium ?

(K_{sp} for AgCl = 1.8×10^{-10}, K_{sp} for $PbCl_2$ = 1.7×10^{-5})

(A) $[Ag^+] = 1.8 \times 10^{-7}$ M, $[Pb^{2+}] = 1.7 \times 10^{-6}$ M

(B) $[Ag^+] = 1.8 \times 10^{-11}$ M, $[Pb^{2+}] = 8.5 \times 10^{-5}$ M

(C) $[Ag^+] = 1.8 \times 10^{-9}$ M, $[Pb^{2+}] = 1.7 \times 10^{-3}$ M

(D) $[Ag^+] = 1.8 \times 10^{-11}$ M, $[Pb^{2+}] = 1.7 \times 10^{-4}$ M

Open solution

Q43. AIPMT 2011 Prelims

A buffer solution is prepared in which the concentration of NH_3 is 0.30 M and the concentration of NH^+_4 is 0.20 M. If the equilibrium constant, K_b for NH_3 equals 1.8×10^{-5}, what is the pH of this solution? (log 2.7 = 0.43)

(A) 9.08 (B) 9.43 (C) 11.72 (D) 8.73

Open solution

Q44. AIPMT 2011 Prelims

Which of the following is least likely to behave as Lewis base?

(A) H_2O (B) NH_3 (C) BF_3 (D) OH^-

Open solution

Q45. AIPMT 2010 Prelims

If pH of a saturated solution of $Ba(OH)_2$ is 12, the value of its K_{sp} is

(A) 4.00×10^{-6} M^3 (B) 4.00×10^{-7} M^3

(C) 5.00×10^{-6} M^3 (D) 5.00×10^{-7} M^3

Open solution

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Q46. AIPMT 2010 Prelims

What is $[H^+]$ in mol/L of a solution that is 0.20 M in CH_3COONa and 0.10 M in CH_3COOH? K_a for $CH_3COOH = 1.8 \times 10^{-5}$

(A) 3.5×10^{-4} (B) 1.1×10^{-5} (C) 1.8×10^{-5} (D) 9.0×10^{-6}

Open solution

Q47. AIPMT 2010 Prelims

In a buffer solution containing equal concentration of B^- and HB, the K_b for B^- is 10^{-10}. The pH of buffer solution is

(A) 10 (B) 7 (C) 6 (D) 4

Open solution

Q48. AIPMT 2009

Which of the following molecules acts as a Lewis acid?

(A) $(CH_3)_2O$ (B) $(CH_3)_3P$ (C) $(CH_3)_3N$ (D) $(CH_3)_3B$

Open solution

Q49. AIPMT 2009

The ionization constant of ammonium hydroxide is 1.77×10^{-5} at 298 K. Hydrolysis constant of ammonium chloride is

(A) 6.50×10^{-12} (B) 5.65×10^{-13} (C) 5.65×10^{-12} (D) 5.65×10^{-10}

Open solution

Q50. AIPMT 2009

What is the $[OH^-]$ in the final solution prepared by mixing 20.0 mL of 0.050 M HCl with 30.0 mL of 0.10 M $Ba(OH)_2$?

(A) 0.40 M (B) 0.0050 M (C) 0.12 M (D) 0.10 M

Open solution

Q51. AIPMT 2008

Equal volumes of three acid solutions of pH 3, 4 and 5 are mixed in a vessel. What will be the H^+ ion concentration in the mixture?

(A) 3.7×10^{-3} M (B) 1.11×10^{-3} M (C) 1.11×10^{-4} M (D) 3.7×10^{-4} M

Open solution

Q52. AIPMT 2007

A weak acid, HA, has a K_a of 1.00×10^{-5}. If 0.100 mol of this acid is dissolved in one litre of water, the percentage of acid dissociated at equilibrium is closest to

(A) 1.00% (B) 99.9% (C) 0.100% (D) 99.0%

Open solution

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Q53. AIPMT 2007

Calculate the pOH of a solution at 25°C that contains 1×10^{-10} M of hydronium ions, i.e. H_3O^+.

(A) 4.000 (B) 9.000 (C) 1.000 (D) 7.000

Open solution

Q54. AIPMT 2006

Which of the following pairs constitutes a buffer?

(A) HCl and KCl (B) HNO_2 and $NaNO_2$

(C) NaOH and NaCl (D) HNO_3 and NH_4NO_3

Open solution

Q55. AIPMT 2006

The hydrogen ion concentration of a 10^{-8} M, HCl aqueous solution at 298 K ($K_w = 10^{-14}$) is

(A) 1.0×10^{-8} M (B) 1.0×10^{-6} M

(C) 1.0525×10^{-7} M (D) 9.525×10^{-8} M

Open solution

Q56. AIPMT 2005

At 25°C, the dissociation constant of a base, BOH, is 1.0×10^{-12}. The concentration of hydroxyl ions in 0.01 M aqueous solution of the base would be

(A) 1.0×10^{-5} mol L^{-1} (B) 1.0×10^{-6} mol L^{-1}

(C) 2.0×10^{-6} mol L^{-1} (D) 1.0×10^{-7} mol L^{-1}

Open solution

Q57. AIPMT 2005

H_2S gas when passed through a solution of cations containing HCl precipitates the cations of second group of qualitative analysis but not those belonging to the fourth group. It is because

(A) presence of HCl decreases the sulphide ion concentration

(B) solubility product of group II sulphides is more than that of group IV sulphates

(C) presence of HCl increases the sulphide ion concentration

(D) sulphides of group IV cations are unstable in HCl.

Open solution

Q58. AIPMT 2004

The solubility product of a sparingly soluble salt AX_2 is 3.2×10^{-11}. Its solubility (in moles/L) is

(A) 5.6×10^{-6} (B) 3.1×10^{-4} (C) 2×10^{-4} (D) 4×10^{-4}

Open solution

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Q59. AIPMT 2004

The rapid change of pH near the stoichiometric point of an acid-base titration is the basis of indicator detection. pH of the solution is related to ratio of the concentrations of the conjugate acid (HIn) and base (In^-) forms of the indicator by the expression

(A) $\log \frac{[In^-]}{[HIn]} = pK_{In} - pH$ (B) $\log \frac{[HIn]}{[In^-]} = pK_{In} - pH$

(C) $\log \frac{[HIn]}{[In^-]} = pH - pK_{In}$ (D) $\log \frac{[In^-]}{[HIn]} = pH - pK_{In}$

Open solution

Q60. AIPMT 2003

The solubility product of AgI at 25°C is 1.0×10^{-16} $mol^2\ L^{-2}$. The solubility of AgI in 10^{-4} N solution of KI at 25°C is approximately (in mol L^{-1})

(A) 1.0×10^{-16} (B) 1.0×10^{-12} (C) 1.0×10^{-10} (D) 1.0×10^{-8}

Open solution

Q61. AIPMT 2002

Solubility of MX_2 type electrolytes is 0.5×10^{-4} mole/lit., then find out K_{sp} of electrolytes.

(A) 5×10^{-12} (B) 25×10^{-10} (C) 1×10^{-13} (D) 5×10^{-13}

Open solution

Q62. AIPMT 2002

Which has highest pH?

(A) CH_3COOK (B) Na_2CO_3 (C) NH_4Cl (D) $NaNO_3$

Open solution

Q63. AIPMT 2002

Solution of 0.1 N NH_4OH and 0.1 N NH_4Cl has pH 9.25. Then find out pK_b of NH_4OH

(A) 9.25 (B) 4.75 (C) 3.75 (D) 8.25

Open solution

Q64. AIPMT 2001

In HS^-, I^-, $R - NH_2$, NH_3 order of proton accepting tendency will be

(A) $I^- > NH_3 > R - NH_2 > HS^-$ (B) $NH_3 > R - NH_2 > HS^- > I^-$

(C) $R - NH_2 > NH_3 > HS^- > I^-$ (D) $HS^- > R - NH_2 > NH_3 > I^-$

Open solution

Q65. AIPMT 2001

Ionisation constant of CH_3COOH is 1.7×10^{-5} and concentration of H^+ ions is 3.4×10^{-4}. Then find out initial concentration of CH_3COOH molecules.

(A) 3.4×10^{-4} (B) 3.4×10^{-3} (C) 6.8×10^{-4} (D) 6.8×10^{-3}

Open solution

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Q66. AIPMT 2001

Solubility of M_2S salt is 3.5×10^{-6} then find out solubility product.

(A) 1.7×10^{-6} (B) 1.7×10^{-16} (C) 1.7×10^{-18} (D) 1.7×10^{-12}

Open solution

Q67. AIPMT 2000

Correct relation between dissociation constants of a dibasic acid is

(A) $K_{a_1} = K_{a_2}$ (B) $K_{a_1} > K_{a_2}$

(C) $K_{a_1} < K_{a_2}$ (D) $K_{a_1} = \frac{1}{K_{a_2}}$

Open solution

Q68. AIPMT 2000

Conjugate acid of NH_2^- is

(A) NH_4OH (B) NH_4^+ (C) NH_2^- (D) NH_3

Open solution

Q69. AIPMT 2000

Which statement is wrong about pH and H^+?

(A) pH of neutral water is not zero.

(B) Adding 1 N solution of CH_3COOH and 1 N solution of NaOH, pH will be seven.

(C) $[H^+]$ of dilute and hot H_2SO_4 is more than concentrated and cold H_2SO_4

(D) Mixing solution of CH_3COOH and HCl. pH will be less than 7.

Open solution

Q70. AIPMT 1999

The strongest conjugate base is

(A) NO_3^- (B) Cl^- (C) SO_4^{2-} (D) CH_3COO^-

Open solution

Q71. AIPMT 1999

The concentration of $[H^+]$ and concentration of $[OH^-]$ of a $0.1\ M$ aqueous solution of 2% ionised weak monobasic acid is

[ionic product of water $= 1 \times 10^{-14}$]

(A) $0.02 \times 10^{-3}\ M$ and $5 \times 10^{-11}\ M$ (B) $1 \times 10^{-3}\ M$ and $3 \times 10^{-11}\ M$

(C) $2 \times 10^{-3}\ M$ and $5 \times 10^{-12}\ M$ (D) $3 \times 10^{-2}\ M$ and $4 \times 10^{-13}\ M$

Open solution

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Q72. AIPMT 1999

The solubility of a saturated solution of calcium fluoride is $2 \times 10^{-4}\ \mathrm{mol/L}$. Its solubility product is

(A) 12×10^{-2} (B) 14×10^{-4} (C) 22×10^{-11} (D) 32×10^{-12}

Open solution

Q73. AIPMT 1997

A physician wishes to prepare a buffer solution at $\mathrm{pH} = 3.85$ that efficiently resists changes in pH yet contains only small concentration of the buffering agents. Which of the following weak acids together with its sodium salt would be best to use?

(A) 2, 5-Dihydroxy benzoic acid ($\mathrm{p}K_a = 2.97$) (B) Acetoacetic acid ($\mathrm{p}K_a = 3.58$)

(C) *m*-Chlorobenzoic acid ($\mathrm{p}K_a = 3.98$) (D) *p*-Chlorocinnamic acid ($\mathrm{p}K_a = 4.41$)

Open solution

Q74. AIPMT 1997

The solubility product of CuS, $\mathrm{Ag_2S}$ and HgS are 10^{-31}, 10^{-44} and 10^{-54} respectively. The solubilities of these sulphides are in the order

(A) $\mathrm{HgS} > \mathrm{Ag_2S} > \mathrm{CuS}$ (B) $\mathrm{CuS} > \mathrm{Ag_2S} > \mathrm{HgS}$

(C) $\mathrm{Ag_2S} > \mathrm{CuS} > \mathrm{HgS}$ (D) $\mathrm{AgS} > \mathrm{HgS} > \mathrm{CuS}$

Open solution

Q75. AIPMT 1996

The ionic product of water at $25^\circ\mathrm{C}$ is 10^{-14}. Its ionic product at $90^\circ\mathrm{C}$ will be,

(A) 1×10^{-14} (B) 1×10^{-16} (C) 1×10^{-20} (D) 1×10^{-12}

Open solution

Q76. AIPMT 1996

The pH value of $N/10$ NaOH solution is

(A) 12 (B) 13 (C) 10 (D) 11

Open solution

Q77. AIPMT 1996

Which of the following is not a Lewis acid?

(A) $\mathrm{SiF_4}$ (B) $\mathrm{C_2H_4}$ (C) $\mathrm{BF_3}$ (D) $\mathrm{FeCl_3}$

Open solution

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Q78. AIPMT 1995

The pH value of blood does not appreciably change by a small addition of an acid or a base, because the blood

(A) can be easily coagulated

(B) contains iron as a part of the molecule

(C) is a body fluid

(D) contains serum protein which acts as buffer.

Open solution

Q79. AIPMT 1995

The pH value of a $10\ \mathrm{M}$ solution of HCl is

(A) equal to 1 (B) equal to 2 (C) less than 0 (D) equal to 0

Open solution

Q80. AIPMT 1995

The solubility of AgCl will be minimum in

(A) $0.01\ \mathrm{M}\ \mathrm{CaCl_2}$

(B) pure water

(C) $0.001\ \mathrm{M}\ \mathrm{AgNO_3}$

(D) $0.01\ \mathrm{M}\ \mathrm{NaCl}$

Open solution

Q81. AIPMT 1994

Which one of the following is most soluble?

(A) $\mathrm{Bi_2S_3}$ ($K_{sp} = 1 \times 10^{-70}$)

(B) $\mathrm{Ag_2S}$ ($K_{sp} = 6 \times 10^{-51}$)

(C) CuS ($K_{sp} = 8 \times 10^{-37}$)

(D) MnS ($K_{sp} = 7 \times 10^{-16}$)

Open solution

Q82. AIPMT 1994

At $80^\circ\mathrm{C}$, distilled water has $[\mathrm{H_3O^+}]$ concentration equal to 1×10^{-6} mole/litre. The value of K_w at this temperature will be

(A) 1×10^{-12} (B) 1×10^{-15} (C) 1×10^{-6} (D) 1×10^{-9}

Open solution

Q83. AIPMT 1992

0.1 M solution of which one of these substances will act basic?

(A) Sodium borate

(B) Ammonium chloride

(C) Calcium nitrate

(D) Sodium sulphate

Open solution

Q84. AIPMT 1991

Aqueous solution of acetic acid contains

(A) $\mathrm{CH_3COO^-}$ and $\mathrm{H^+}$

(B) $\mathrm{CH_3COO^-}$, $\mathrm{H_3O^+}$ and $\mathrm{CH_3COOH}$

(C) $\mathrm{CH_3COO^-}$, $\mathrm{H_3O^+}$ and $\mathrm{H^+}$

(D) $\mathrm{CH_3COOH}$, $\mathrm{CH_3COO^-}$ and $\mathrm{H^+}$

Open solution

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Q85. AIPMT 1991

Aqueous solution of ammonia consists of

(A) H^+

(B) OH^-

(C) NH_4^+

(D) NH_4^+ and OH^-.

Open solution

Q86. AIPMT 1988

The compound whose water solution has the highest pH is

(A) NaCl

(B) $NaHCO_3$

(C) Na_2CO_3

(D) NH_4Cl

Open solution

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Answer Key

1. **A**	2. **D**	3. **B**	4. **D**	5. **A**
6. **B**	7. **C**	8. **A**	9. **A**	10. **D**
11. **A**	12. **D**	13. **A**	14. **B**	15. **D**
16. **D**	17. **D**	18. **D**	19. **A**	20. **C**
21. **A**	22. **D**	23. **A**	24. **C**	25. **B**
26. **B**	27. **B**	28. **D**	29. **D**	30. **D**
31. **B**	32. **C**	33. **B**	34. **D**	35. **D**
36. **B**	37. **A**	38. **A**	39. **B**	40. **A**
41. **A**	42. **C**	43. **B**	44. **C**	45. **D**
46. **D**	47. **D**	48. **D**	49. **D**	50. **D**
51. **D**	52. **A**	53. **A**	54. **B**	55. **C**
56. **D**	57. **A**	58. **C**	59. **D**	60. **B**
61. **D**	62. **B**	63. **B**	64. **C**	65. **D**
66. **B**	67. **B**	68. **D**	69. **B**	70. **D**
71. **C**	72. **D**	73. **C**	74. **C**	75. **D**
76. **B**	77. **B**	78. **D**	79. **C**	80. **A**
81. **D**	82. **A**	83. **A**	84. **B**	85. **D**
86. **C**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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